Unified Health ID: Overcoming Fragmented Records Low ABHA I'd Adoption

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***Abstract*—India has a rapidly growing burden of chronic conditions like diabetes, hypertension, cardiovascular diseases, and cancers that together are responsible for a large share of mortality in the country. In this regard, initiatives such as the Ayushman Bharat Digital Mission and the ABHA health ID have been launched to improve healthcare. delivery. They have limited impact because of persistent problems in fragmented medical records, low digital penetration in rural and semi-urban areas, and the lack of integrated frameworks supporting long-term recovery. and continuity of care. These challenges lead to multiple systemic problems. Patients are subjected to repeated tests, delays in treatment, increased costs with no participation in managing their own health. Clinically, doctors do not have access to complete longitudinal patient histories; researchers cannot obtain structured anonymized datasets for meaningful medical analysis. To fill these gaps, this project proposes a patient-centric digital health identification system that will consolidate lifelong medical records starting from birth. The system will provide secure consent-based data sharing and personalized evidence-informed lifestyle and natural care recommendations integrated with conventional treatment pathways. A web-based patient interface and clinician dashboard are proposed here to describe the functionality and usability of the system. The continuity of care, reduction of redundancy in diagnostics and costs, increase in patient engagement through personalized insights as well as reliable data for medical research and policy formulation are expected outcomes. This project will help build resilience by addressing such challenges toward an integrated digital health ecosystem in India.**

***Index Terms*—Digital Health, Non-Communicable Diseases (NCDs), Cancer, Telemedicine, Unified health ID.**

I. Introduction

Evidence from national and international health assessments has shown that non-communicable diseases (NCDs) like diabetes, cardiovascular disorders, and cancer are major contributors to premature mortality in India [1], [2]. These are not acute infectious diseases, but chronic ailments that take a long time to form based on changes in lifestyle and exploitation of the environment, as well as the aging of the population [3], [4]. These conditions require long-term, multi-faceted modes of care because they remain with the individual for the better part of their lifetime. The recent surveillance data and policy discussions had placed greater emphasis on changing the health system priorities from an infection control approach to a sustained, ongoing management of noncommunicable diseases in response to the changing patterns of disease in the population. This paper is based on three main pieces of evidence: cancer incidence and prevalence data from ICMR 2025 registry synthesis covering 43 Population-Based and Hospital-Based Cancer Registries (PBCRs/HBCRs); mortality patterns with trend estimates as reported in global burden of disease studies; and programmatic performance documentation from national non-communicable disease programs [5]–[7]. When viewed together, these data sources provide age-adjusted incidence indicators, lifetime risk estimates, and program linkage measures underlining the need for maintenance of longitudinal health records and also continuity in treatment. Mortality Data: This trend analysis is informed by results from the 2025 Lancet study [1], [2]. It was a global assessment that reviewed mortality data across 185 countries for the years between 2001 and 2019. The assessment quantified changes in the probability of death due to non-communicable diseases between birth and 80 years of age with specific reference to India. Policy and Performance Data: The government response and programme implementation will be assessed using progress reports for the National Programme for Prevention and Control of Cancer Diabetes Cardiovascular Diseases and Stroke (NPCDCS/NP-NCD) provided by the Ministry of Health and Family Welfare (MoHFW) as of August 2025 [3]. These reports give cumulative information about volumes of screening done, outcomes of diagnosis, and linkages to treatment for five priority non-communicable diseases hypertension, diabetes, oral cancer breast cancer cervix cancer among people aged over 30 years with service delivery mainly through Ayushman Arogya Mandirs (AAMs).

II. METHODOLOGY

The major limitations discovered in the preliminary analysis were high cancer Mortality-to-Incidence Ratio (MIR) [8], fragmented health records, and low public uptake of existing digital health initiatives. This describes a need for an integrated and patient-centred digital health infrastructure. Hence, this paper presents a proposal to design and develop a comprehensive three-tier web-based application that aims to keep continuous longitudinal health data records for individuals starting from birth.

Architectural Foundation: Unified Health Identifier and Lifelong Health Data Framework The proposed methodology

is based on the issuance of a Unified Health Identifier at birth so that it would be possible to create a continuous lifelong health record. Early assignment of the identifier enables immunization, neonatal, and growth-related data to be captured in one longitudinal framework thus reducing fragmentation of data from inception. This is in line with existing guidance regarding national health identifiers which are recognized as key components necessary for continuity of care at the population level as well as being consistent with internationally accepted practices regarding persistent patient identification.

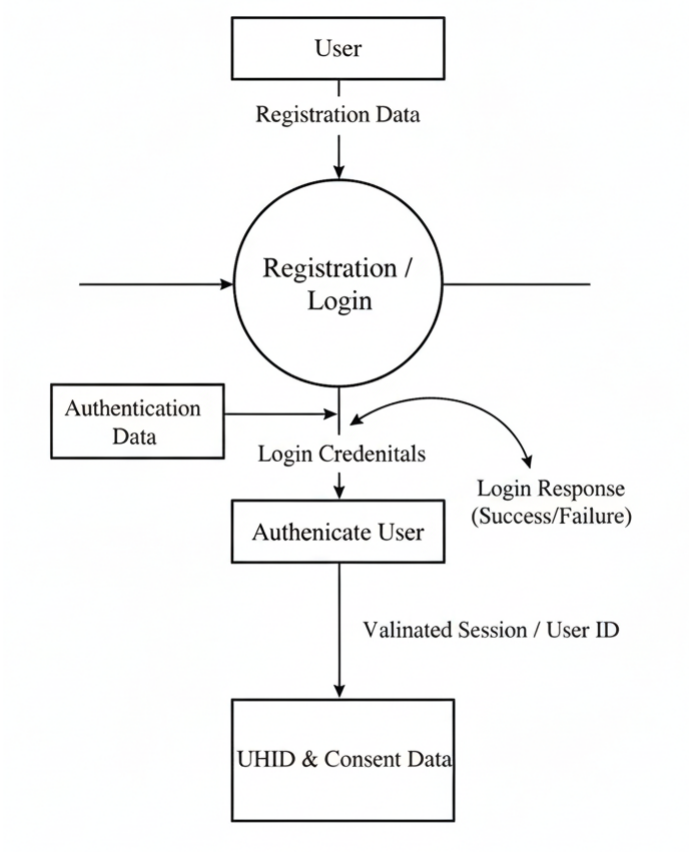

Fig. 1: Data Flow Diagram

Data flow diagram illustrates the user registration and login functionality of the system. The users enter the registration or login details, which are validated by the authentication module. Depending on the validation, the system produces a success or failure message. After successful validation, a validated session or user ID is generated, enabling secure access to UHID and consent information.

Activity diagram shows how patients interact with the system and admin during registration and login operations. The system conducts validation on the registration form which the patient completes before the admin performs the verification process. The system generates a UHID and establishes the patient account after the verification process succeeds. The patient needs to enter their login details which the system verifies to finish the process.

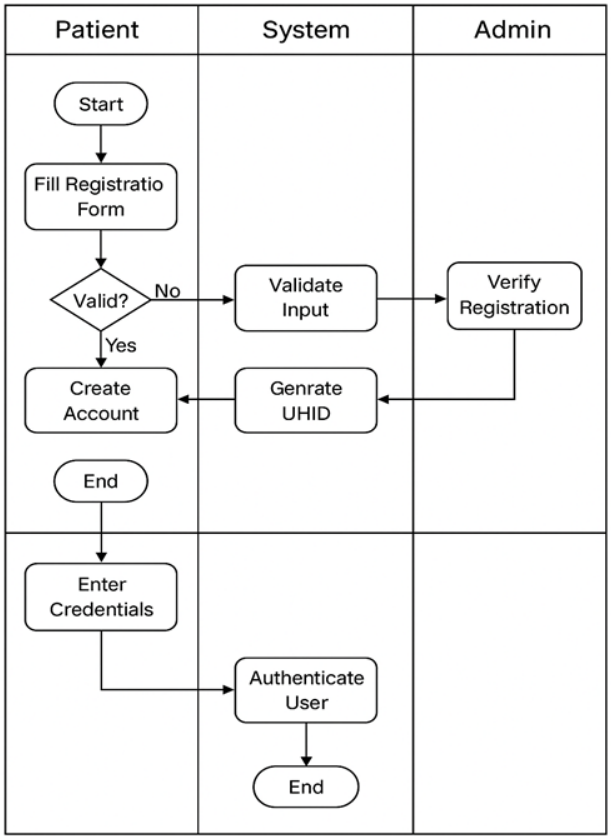

Fig. 2: Activity Diagram

The integrated digital system eases an iterative process for the systematic exchange of information through email, fax, telephone, and face-to-face communication. This level of coordinated communication among patients, clinicians, and public health decision-makers is a major departure from the conventional paper-based documentation system. The digital system allows patients and clinicians to access health information at any time.

III. Results

A look at recent epidemiological data from 2019 to 2025, as well as the national program performance, reveals a shocking contrast: a huge number of people suffering from the disease, on one side, and very high death rates, on the other, all this occurring with considerable system-level implementation gaps.

A. Worsening NCD and Cancer Outcome Indicators

- **Global Mortality Difference:** India is a global health anomaly in that it is the only country that has shown an increase in the probability of dying prematurely from noncommunicable diseases (NCDs) between 2010 and 2019, whereas the trend in other countries has been declining [1], [2]. This negative trend has mainly been caused by heart disease and diabetes conditions [2].
- **Gender Risk Disparity:** The increase in NCD mortality probability was substantially worse for women, with

an increase of 2.1 per cent, while men's increase was marginal at about 0.1 per cent [1].

- **Cancer Survival Indicator:** India's cancer Mortality-to-Incidence Ratio (MIR) recorded at 64.47

The change between these two situations indicates that while the public is generally in favor of a digital health ecosystem, the current ways of communication and operation have failed to convey the fundamentals of digital health. Therefore, a standardized approach to digital health literacy is essential for its widespread and effective adoption.

B. Digital Health Adoption and Awareness Gap

Awareness Gap: Survey results reveal a significant hole in digital health literacy among young adults. Only 37.50% of respondents were fully aware of the purpose and function of the ABHA ID, while the majority, 46.40 %, had partial knowledge; 16.04 % were completely unaware. This gap indicates that despite large-scale national initiatives, the communication strategy about the mechanics and utility of digital health accounts is not effectively reaching the youth, who are often perceived as early adopters of technology. The survey indicates that awareness among Indian youth about this issue is very low.

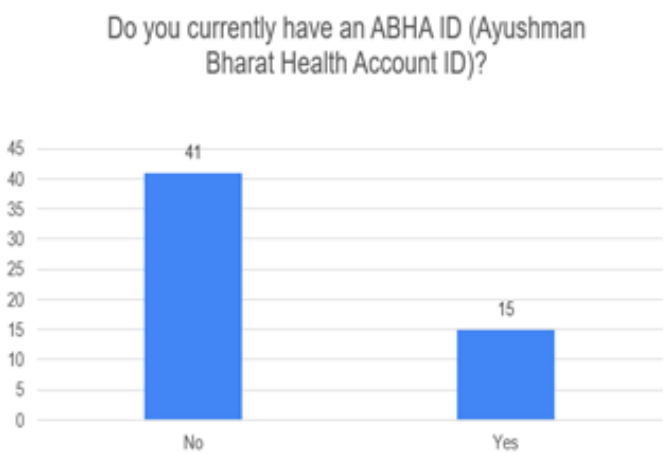

Fig. 3: Survey responses on whether participants have an ABHA ID

Only 15 out of 56 have an ABHA ID that is about 25%; it is depressing to see least awareness amongst the youth. To understand whether the youth have a little knowledge about this cause and the need of such an initiative another survey question and its stats are as follows:

Survey data of young adults aged 16 to 25 indicate a significant difference in the adoption of digital health initiatives:

- **High Optimism:** A combined 73.21% of respondents either 'Agree' (35.71%) or 'Strongly Agree' (37.50%) that the ABHA ID will improve healthcare access and services.
- **Limited Knowledge:** Only 37.5% of those surveyed said that they understood the ABHA ID concept and function within the larger system, the main unit, after the positive anticipation level. The main unit had only a partial awareness (46.4%).

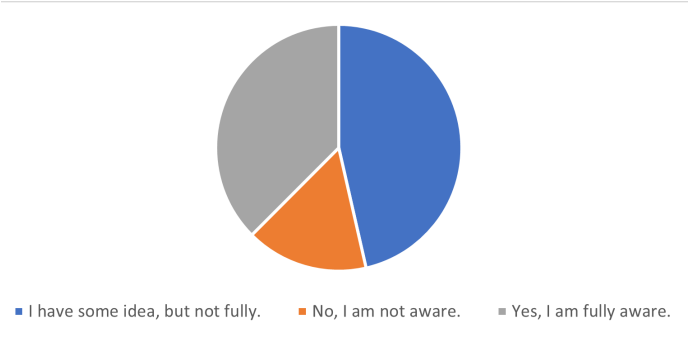

Fig. 4: ABHA ID awareness levels among young adults.

TABLE I: ABHA ID Awareness Among Young Adults (Age 16–25, N=56)

Are you aware of what an ABHA ID is and its purpose?	**Numbers**	**Percentage**
I have some idea, but not fully.	26	46.40
No, I am not aware.	9	16.04
Yes, I am fully aware.	21	37.50

The change between these two situations indicates that while the public is generally in favor of a digital health ecosystem, the current ways of Communication and operation have failed to get the basics of digital health. The standard of digital health literacy should be there if one is to use it widely and effectively.

TABLE II: Public Opinion on ABHA ID Improving Healthcare Access (Age 16–25, N=56)

Opinion	**Numbers**	**Percentage**
Agree	20	35.71
Neutral	15	26.78
Strongly Agree	21	37.50

IV. DISCUSSION

These findings show that a big gap exists between the public health initiatives that India is currently implementing and the measures that should be implemented according to the NP-NCD program. Although a huge number of people have been screened [3], the anticipated decrease in disease and deaths from non-communicable diseases (NCDs) has not been experienced [1], [8]. Fundamentally, the reasons involve the difficulty of turning plans into real actions, the continuation of gender-related social and economic inequalities, and a serious lack of digital health systems.

A. Systematic Failure in Treatment Linkage

- On top of that, the extremely high Cancer Mortality-to-Incidence Ratio of 64.47% [8] is a clear indication of systemic failures post-diagnosis that are linked to

operational challenges at both micro and mesa levels of the NP-NCD framework. This high MIR, when compared to the diabetes linkage gap, is a sign of failure in the post-diagnosis stage—specifically the operations and resources at the facilities and program levels [12]. When the clinical staff posts are not filled, patients do not get the medicines they need regularly and are also poorly followed up with, they end up being non-adherent or defaulting due to various reasons, hence resulting in a high mortality burden [13], [15], [16]. Therefore, it is critical that these implementation bottlenecks [3] are addressed if we are to convert the benefits gained from the screening into actually improved survival.

- **T**he implementation of the NPCDCS [12]–[14] monitoring framework remains a challenge (as evidenced by the incomplete execution), and this, along with the poor quality of the health data at the national level in India, results in the basket of policy execution being thrown away [15]. The problem is that such a data gap causes the identification of blocks to be difficult and, at the same time, it results in the continuation of misreporting [14]. Besides that, poor internet connectivity in remote areas coupled with software that is not very user-friendly means that digital reporting mechanisms used for capturing structured data on NCD risk factors - are compromised, even while at the same time, the national efforts to increase surveillance coverage are on the rise.

B. The Necessity of a Unified Digital System

Speaking of addressing the gaps in data interoperability, patient de-duplication and rural access, the digital health platform proposed in the Methodology takes note of the existing problems and sets out to solve them [24]–[26].

- **C**oming from the surveys, there's a lot of faith in the power of digitisation, but low digital literacy underlines the need for something as fundamental as a unified health identifier to be given to newborns, so that we can get over the low adoption rate and the technical hurdles of current initiatives. ·It is very significant that the local community and family doctors, especially those from Primary Health Centers, offer support and encourage people to register for such initiatives [27].
- If we employ a three-pronged approach, we can establish a lifelong health record starting from birth, thus eliminating the fragmentation that limits us in chronic disease management. Should we fail to do so, our health care providers will not have the means to access the full medical history of the patient which is basically indispensable for the treatment and saving the costs that unnecessarily happen.
- **T**he Central Administration Panel will be able to respond to the NPCDCS surveillance challenges in a more efficient manner through a platform that features a system capable of securely pooling anonymized data in a single location and thus giving policymakers the ability to have a fully comprehensive, real-time data which can reveal them the level of different intervention successes and also the areas where more need to be applied in order to fix the health inequities at the regional level [12], [28].

V. FUTURE SCOPE AND DEVELOPMENT

If we look at the next 50 years of healthcare in India, the system that we will be having is going to be very predictive and personalized focusing heavily on the community and all this supported by a highly secure and efficient digital infrastructure. The system that we are talking about is better known as the Unified Health Identifier or UHID and it essentially opens the door for us to initiate this transformation and at the same time make sure that even the poorest and disadvantaged sections of the society get health care access.

A. Enhancing Inclusivity: Addressing Digital Literacy Gaps and Connecting the Rural Divide

When confronted with the problem of digital literacy, it is imperative that the UHID should not become a tool that only English speakers and tech-savvy people benefit from. We shall handle non-technical interfaces very soon through the use of the existing network of Accredited Social Health Activists and Auxiliary Nurse Midwives who will be the locals' first point of contact with the healthcare system. These health workers at the frontline will be using a Clinician Dashboard that through voice-activated commands and regionally supported language tools will enable them to input and retrieve patient data. At the same time, the family members who in a way will be the social force behind this implementation will be assisting these workers. Mid-way, the system can become more elaborate by including top-notch telemedicine services, which means having a nationwide digital infrastructure that is supported by well-explained public awareness campaigns that are aimed at people learning how to make use of the system and also becoming more digitally literate. Specialist care can be taken to rural areas by this means, and what is more, it helps women who usually suffer the most in healthcare access issues. Eventually, the UHID framework will be able to unify voice and regional biometric authentication, thereby doing away with complicated passwords and the need for language-preference interfaces when it comes to the most important health records.

B. Future Development of the Unified Health System

Future project development should focus on system integration with the medical technology ecosystem which is constantly growing:

- **I**t's very important that the project continue by steadily integrating the system with the rapidly expanding medical technology ecosystem. This may be achieved by introducing Clinical Decision Support Systems that would integrate AI into Clinician Dashboard and provide doctors and physicians with instant, evidence-based decision-making guidance.
- **U**HID should, from patient's perspective, add open and secure APIs that will easily connect the data from patient-owned wearables and remote monitoring devices. This

will enable us to monitor the conditions of chronic patients discreetly and provide the patients with personalized and evidence-based advice aligned to Ayurvedic, Yoga and Naturopathy teachings.

- **H**ence, the UHID system is not only a digitization effort; it is a national platform for enabling equitable and advanced health management that will cover the whole socio-economic spectrum of India in the next fifty years.

VI. CONCLUSION

As far as epidemiological trends in India are concerned, what is most eye-catching is the recent sharp rise in deaths due to non-communicable diseases [1], in fact, it is one of the most shocking in the world. The cancer Mortality-to Incidence Ratio in India is 64.47% [8] which is a very discouraging figure, and a big difference from the scenario that poor screening [3] can simply cause, as it is in this emergency [9], [22].

The crisis derives from two main sources: the failure to link, monitor and ensure a continuity of care. On the one hand, there are the operational issues that have resulted in the state of the system being characterized by unoccupied doctor positions, and irregular drug supplies. This is really ambitious vision that we cannot deliver in its whole without a total reorganization of the way healthcare is delivered to us in a fragmented manner and creating a complete digital health system that would be made up of different plans and functional objectives that are intertwined.

The blueprint for the digital health system will revolve around the introduction of a Unified Health ID which will be supported by integrated Patient and Clinician Portals.

- **U**niversal longitudinal health records of all humans going back to birth [18].
- **T**he continuation of informed care; in this way, the cancer mortality-incidence ratio will be reduced while treatment adherence (especially for diabetics) will be improved.
- **D**elivering accurate real-time data to National Policy Makers that will help them to develop, implement and evaluate timely targeted interventions against Regional and Systemic Health Disparities.

As mentioned, the digital health system thus placed out will not only be a requirement for India to attain its sustainable development goals but will also be regarded as a planned weapon for transitioning from mere disease incidence surveillance and tracking to actually saving lives in the pursuit of sustainable development goals in India.

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